

Chapter 8: Spin 1 and Gauge Invariance

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8.1 Unitary Representations of the Poincaré Group

Poincaré Group $ISO(1, 3)$

The symmetry group of Minkowski space consists of spacetime translations and Lorentz transformations:

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} + a^{\mu}$$

- a^{μ} : Constant 4-vector representing spacetime translation.
- Λ^{μ}_{ν} : Real 4×4 matrix satisfying $\Lambda^{\mu}_{\alpha} \Lambda^{\nu}_{\beta} g_{\mu\nu} = g_{\alpha\beta}$ (Lorentz transformation).
- **Physical Particles**: Defined as **irreducible unitary representations** of the Poincaré group, uniquely classified by mass m and spin J (Wigner, 1939).

8.1.1 Unitarity versus Lorentz Invariance

Non-Relativistic Quantum Mechanics

States $|\psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ have a positive-definite norm:

$$\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2 > 0$$

- Preserved under $SU(2)$ rotation matrices (δ_{ij} metric).

4-Vector State in Relativity

A 4-vector state $|\psi\rangle = c_0|V^0\rangle + c_1|V^1\rangle + c_2|V^2\rangle + c_3|V^3\rangle$ has the Lorentz-invariant inner product:

$$\langle\psi|\psi\rangle = g_{\mu\nu}c^{*\mu}c^\nu = -|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2$$

- $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$: Minkowski metric.
- c_0 : Time-component coefficient leading to a **negative norm (ghost state)**.
- **Conflict**: Unitarity requires positive-definite norms ($\delta_{\mu\nu}$), while relativity requires Lorentz invariant norms ($g_{\mu\nu}$).

8.2 Embedding Particles into Fields

Energy Density and Positivity

To prevent runaway instabilities in classical field theory, the energy density must be bounded from below:

$$\mathcal{E} = T^{00} = \sum_n \frac{\partial \mathcal{L}}{\partial \dot{\phi}_n} \dot{\phi}_n - \mathcal{L} \geq 0$$

- T^{00} : Time-time component of the energy-momentum tensor $T^{\mu\nu}$.
- ϕ_n : Generic field components.
- $\dot{\phi}_n \equiv \frac{\partial \phi_n}{\partial t}$: Time derivative of the field.

8.2.1 Spin 0

Real Scalar Field Lagrangian

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

- $\phi(x)$: Real scalar field (1 physical degree of freedom).
- m : Particle mass.
- $\partial_\mu \equiv \frac{\partial}{\partial x^\mu}$: Spacetime derivative.

Equation of Motion & Energy Density

$$(\square + m^2)\phi = 0$$

$$\mathcal{E} = \frac{1}{2} \left[(\partial_t \phi)^2 + (\vec{\nabla} \phi)^2 + m^2 \phi^2 \right] \geq 0$$

- $\square \equiv \partial_\mu \partial^\mu = \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2$: d'Alembertian operator.
- $\mathcal{E} \geq 0$: Bounded from below, ensuring a stable physical vacuum.

8.2.2 Massive Spin 1 (1/3)

General 2nd-Derivative Lagrangian for Vector Field A_μ

$$\mathcal{L} = a A_\mu \square A^\mu + b A_\mu \partial^\mu \partial^\nu A_\nu + m^2 A_\mu A^\mu$$

- $A_\mu(x) = (A_0, \vec{A})$: 4-vector field (4 components).
- a, b : Arbitrary dimensionless real coefficients to be determined.

Equations of Motion

$$2a \square A_\mu + 2b \partial_\mu (\partial_\nu A^\nu) + 2m^2 A_\mu = 0$$

Taking the 4-divergence ∂^μ on both sides:

$$\partial^\mu [a \square A_\mu + b \partial_\mu (\partial_\nu A^\nu) + m^2 A_\mu] = 0 \implies [(a + b) \square + m^2] (\partial_\mu A^\mu) = 0$$

8.2.2 Massive Spin 1 (2/3)

Eliminating the Spin-0 Ghost Component

To convert $\partial_\mu A^\mu$ from a propagating dynamical field into an algebraic constraint, we must set $a = -b$:

$$m^2(\partial_\mu A^\mu) = 0 \implies \partial_\mu A^\mu = 0 \quad (\text{Lorenz Constraint})$$

Proca Lagrangian

Choosing the standard normalization $a = 1/2, b = -1/2$:

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu$$

- $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$: Anti-symmetric field strength tensor.
- Equations of Motion:

$$(\square + m^2)A_\mu = 0 \quad \text{and} \quad \partial_\mu A^\mu = 0$$

8.2.2 Massive Spin 1 (3/3)

Polarization Vectors

For momentum $p^\mu = (E, 0, 0, p_z)$ with $p^2 = E^2 - p_z^2 = m^2$:

- **Transverse Polarizations:**

$$\epsilon_1^\mu = (0, 1, 0, 0), \quad \epsilon_2^\mu = (0, 0, 1, 0)$$

- **Longitudinal Polarization:**

$$\epsilon_L^\mu = \left(\frac{p_z}{m}, 0, 0, \frac{E}{m} \right)$$

Constraints and Normalization

$$p_\mu \epsilon_j^\mu(p) = 0, \quad \epsilon_{j,\mu} \epsilon_k^{*\mu} = -\delta_{jk} \quad (j, k \in \{1, 2, L\})$$

- $E = \sqrt{p_z^2 + m^2}$: On-shell particle energy.
- m : Rest mass (ensures longitudinal polarization is well-defined).
- Physical degrees of freedom: $4 - 1 = 3$.

8.2.3 Massless Spin 1

Maxwell Lagrangian ($m \rightarrow 0$ Limit)

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

- Equations of Motion: $\partial_\mu F^{\mu\nu} = \square A^\nu - \partial^\nu(\partial_\mu A^\mu) = 0$.

Gauge Invariance

$\mathcal{L}$ is invariant under $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu\alpha(x)$

- $\alpha(x)$: Arbitrary smooth scalar gauge function.

Coulomb Gauge ($\vec{\nabla} \cdot \vec{A} = 0$)

- $\mu = 0$ equation: $\vec{\nabla}^2 A_0 = 0 \implies A_0 = 0$ (nondynamical constraint).
- Physical degrees of freedom: $4 - 1$ (gauge condition) $- 1$ ($A_0 = 0$ constraint) = **2** transverse polarizations ($\epsilon_1^\mu, \epsilon_2^\mu$).

8.2.4 Summary of Spin-1 Representations

Mass State	Lagrangian	Gauge Symmetry	Little Group	Physical Degrees of Freedom
Massive ($m > 0$)	$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}F_{\mu\nu}^2 + \frac{1}{2}m^2 A_\mu^2$	None	$SO(3)$	3 ($\epsilon_1, \epsilon_2, \epsilon_L$)
Massless ($m = 0$)	$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}^2$	$A_\mu \rightarrow A_\mu + \partial_\mu \alpha$	$ISO(2)$	2 (ϵ_1, ϵ_2)

- **Little Group:** The subgroup of Lorentz transformations leaving the reference 4-momentum p^μ invariant (Wigner's classification).
 - Massive ($p^\mu = (m, 0, 0, 0)$): Rotation group $SO(3) \implies 2J + 1 = 3$.
 - Massless ($p^\mu = (E, 0, 0, E)$): Euclidean group of 2D plane $ISO(2) \implies 2$ helicities.

8.3 Covariant Derivatives

Local $U(1)$ Transformation of Complex Scalar Field

$$\phi(x) \rightarrow e^{-iQ\alpha(x)}\phi(x)$$

- $\phi(x) = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$: Complex scalar field.
- Q : Charge generator of the field.
- $\alpha(x)$: Local spacetime-dependent phase parameter.

Definition of Covariant Derivative D_μ

$$D_\mu\phi \equiv (\partial_\mu - ieQA_\mu)\phi$$

- e : Gauge coupling constant (elementary electric charge).
- $A_\mu(x)$: Vector gauge field transforming as $A_\mu \rightarrow A_\mu + \frac{1}{e}\partial_\mu\alpha(x)$.

Transformation property:

$$D_\mu\phi \rightarrow e^{-iQ\alpha(x)}D_\mu\phi$$

8.3.1 Gauge Symmetries and Conserved Currents

Scalar QED Lagrangian

$$\mathcal{L}_{\text{sQED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - m^2\phi^*\phi$$

Expanding the covariant derivative terms:

$$\mathcal{L}_{\text{sQED}} = -\frac{1}{4}F_{\mu\nu}^2 + \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi - eQA_\mu[i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi)] + e^2Q^2A_\mu A^\mu\phi^*\phi$$

Conserved Noether Current J_μ

$$J_\mu = -iQ(\phi^*\partial_\mu\phi - \phi\partial_\mu\phi^*) - 2eQ^2A_\mu\phi^*\phi, \quad \partial^\mu J_\mu = 0$$

- Interaction Lagrangian coupling: $\mathcal{L}_{\text{int}} = -eA_\mu J^\mu$.

8.4 Quantization and the Ward Identity

8.4.1 Massive Spin 1 Quantization

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2\omega_p}} \sum_{j=1}^3 \left(\epsilon_j^\mu(p) a_{\vec{p},j} e^{-ip \cdot x} + \epsilon_j^{*\mu}(p) a_{\vec{p},j}^\dagger e^{ip \cdot x} \right)$$

- $\omega_p = \sqrt{\vec{p}^2 + m^2}$: On-shell energy.
- $a_{\vec{p},j}, a_{\vec{p},j}^\dagger$: Creation and annihilation operators satisfying $[a_{\vec{p},j}, a_{\vec{q},k}^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{q}) \delta_{jk}$.

Positive-Definite Norm

For any state $|\psi\rangle = \sum_{j=1}^3 c_j |p, \epsilon_j\rangle$:

$$\langle\psi|\psi\rangle = |c_1|^2 + |c_2|^2 + |c_3|^2 > 0$$

- Massive spin-1 quantum field theory is fully unitary and Lorentz invariant.

8.4.2 Massless Spin 1 and the Ward Identity (1/2)

Massless Vector Field Expansion

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3 \sqrt{2\omega_p}} \sum_{i=1}^2 \left(\epsilon_i^\mu(p) a_{\vec{p},i} e^{-ip \cdot x} + \epsilon_i^{*\mu}(p) a_{\vec{p},i}^\dagger e^{ip \cdot x} \right)$$

- Sum runs only over the **2 physical transverse polarizations** ($i \in \{1, 2\}$).

Transformation under Massless Little Group $ISO(2)$

Under a general Lorentz transformation Λ , the polarization vector transforms as:

$$\epsilon_\mu(p) \rightarrow c_1 \epsilon_{1\mu}(p) + c_2 \epsilon_{2\mu}(p) + c_3 p_\mu$$

- c_1, c_2, c_3 : Real scalar functions depending on Λ and p^μ .
- p_μ : Photon 4-momentum ($p^2 = 0$).

8.4.2 Massless Spin 1 and the Ward Identity (2/2)

Scattering Amplitude Matrix Element

Let $\mathcal{M} = \epsilon_\mu(p) \mathcal{M}^\mu(p)$ be the scattering amplitude for a process involving an external photon with polarization ϵ_μ .

Under Lorentz transformation:

$$\mathcal{M} \rightarrow c_1 \epsilon_{1\mu} \mathcal{M}'^\mu + c_2 \epsilon_{2\mu} \mathcal{M}'^\mu + c_3 p_\mu \mathcal{M}'^\mu$$

The Ward Identity

For the physical amplitude $\mathcal{M}$ to be Lorentz invariant, the unphysical $c_3 p_\mu$ contribution must vanish identically:

$$p_\mu \mathcal{M}^\mu(p) = 0 \quad (\text{Ward Identity})$$

- **Physical Meaning:** Gauge invariance is the necessary mechanism ensuring that unphysical longitudinal/scalar photon modes decouple, preserving both Unitarity and Lorentz Invariance.

8.5 The Photon Propagator (1/2)

Equations of Motion in Momentum Space

In the presence of an external current $J_\nu(p)$:

$$(-p^2 g_{\mu\nu} + p_\mu p_\nu) A^\mu(p) = J_\nu(p)$$

- Kinetic Operator: $K_{\mu\nu}(p) \equiv -p^2 g_{\mu\nu} + p_\mu p_\nu$.
- Singular Matrix: $K_{\mu\nu} p^\mu = -p^2 p_\nu + p^2 p_\nu = 0 \implies \det K = 0$.
- **Problem:** $K_{\mu\nu}$ has no inverse; the propagator cannot be defined without gauge fixing!

R_ξ Gauge-Fixing Lagrangian

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 - J_\mu A^\mu$$

- ξ : Arbitrary gauge parameter ($\xi \in [0, \infty)$).

8.5 The Photon Propagator (2/2)

Inverting the Modified Kinetic Operator

Modified equation of motion:

$$\left[-p^2 g_{\mu\nu} + \left(1 - \frac{1}{\xi} \right) p_\mu p_\nu \right] A^\nu(p) = J_\mu(p)$$

Inverting the bracketed operator yields the R_ξ **Gauge Photon Propagator**:

$$i\Pi_{\mu\nu}(p) = \frac{-i}{p^2 + i\varepsilon} \left[g_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right]$$

- $\Pi_{\mu\nu}(p)$: Momentum-space photon propagator tensor.
- $+i\varepsilon$: Feynman prescription ensuring time-ordered boundary conditions.

8.5.1 Covariant Gauges

Common Gauge Choices

1. **Feynman–'t Hooft Gauge** ($\xi = 1$):

$$i\Pi_{\mu\nu}(p) = \frac{-ig_{\mu\nu}}{p^2 + i\varepsilon}$$

- Most commonly used in practical calculations due to its simple tensor structure.

2. **Lorenz Gauge** ($\xi = 0$):

$$i\Pi_{\mu\nu}(p) = \frac{-i}{p^2 + i\varepsilon} \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2} \right)$$

- Strictly enforces the Lorenz gauge condition $\partial_\mu A^\mu = 0$ in internal lines.

3. **Gauge Parameter Independence:** Any physical S -matrix amplitude $\mathcal{M}$ is completely independent of ξ because $p_\mu \mathcal{M}^\mu = 0$ (Ward Identity).

8.6 Is Gauge Invariance Real?

Gauge Invariance as Redundancy

- **Physical Symmetries** (e.g., global $U(1)$): Connect distinct physical states, leading to observable conservation laws (e.g., charge conservation).
- **Gauge Invariance**: Not a physical symmetry, but a **mathematical redundancy of description**. It is introduced to describe a massless spin-1 particle (2 physical states) using a local 4-vector field $A_\mu(x)$ (4 components) without violating Lorentz invariance.

The Stueckelberg Model Example

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \frac{1}{2}m^2(A_\mu + \partial_\mu\pi)^2$$

- Gauge transformation: $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$, $\pi \rightarrow \pi - \alpha$.
- Fixing gauge $\pi = 0$ reduces this to the Proca Lagrangian for a single massive spin-1 field with 3 physical degrees of freedom.

8.7 Higher-Spin Fields

8.7.1 Longitudinal Fields and Spin 1

Decompose a vector field into transverse and longitudinal parts:

$$A_\mu = A_\mu^T + \partial_\mu \pi, \quad \text{where } \partial_\mu A^{\mu,T} = 0$$

Substitute into the general vector Lagrangian $\mathcal{L} = a A_\mu \square A^\mu + b A_\mu \partial^\mu \partial^\nu A_\nu + m^2 A_\mu A^\mu$:

$$\mathcal{L} = a A_\mu^T \square A^{\mu,T} + m^2 (A_\mu^T)^2 - (a + b) \pi \square^2 \pi - m^2 \pi \square \pi$$

- **Ghost Analysis:** If $a + b \neq 0$, the k^4 propagator term $\frac{1}{k^4 - m^2 k^2} \sim \frac{1}{k^2} - \frac{1}{k^2 - m^2}$ generates negative-norm ghost states, destroying Unitarity.
- **Uniqueness:** Enforcing $a = -b$ uniquely isolates the physical Proca Lagrangian.

8.7.2 Spin 2 Fields

Massive Spin 2: Fierz-Pauli Lagrangian

Described by a symmetric tensor $h_{\mu\nu}$ (10 components, 5 physical degrees of freedom for $m > 0$):

$$\mathcal{L}_{\text{FP}} = \frac{1}{2} h_{\mu\nu} \square h^{\mu\nu} - h_{\mu\nu} \partial^\mu \partial_\alpha h^{\nu\alpha} + h \partial_\mu \partial_\nu h^{\mu\nu} - \frac{1}{2} h \square h + \frac{1}{2} m^2 (h_{\mu\nu} h^{\mu\nu} - h^2)$$

- $h \equiv h^\mu_\mu = g^{\mu\nu} h_{\mu\nu}$: Trace of the tensor field.
- Uniquely cancels higher-derivative ghost modes for spin-2 particles.

Massless Spin 2 (Graviton)

- 2 physical degrees of freedom. Requires linear gauge symmetry:

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

- Coupling: Must couple to a conserved energy-momentum tensor ($\mathcal{L}_{\text{int}} = -\frac{1}{2} h_{\mu\nu} T^{\mu\nu}$ with $\partial_\mu T^{\mu\nu} = 0$), yielding **General Relativity**.

8.7.3 Spin Greater than 2

Weinberg-Witten Theorem

In any relativistic quantum field theory with a Lorentz-covariant energy-momentum tensor $T^{\mu\nu}$, there cannot exist interacting massless particles with spin $s > 1$ that carry a conserved charge, nor interacting massless particles with spin $s > 2$.

Physical Implications

- **Spin 1:** Massless gauge bosons (photons, gluons) couple to conserved vector currents J^μ .
- **Spin 2:** Massless graviton couples to conserved energy-momentum tensor $T^{\mu\nu}$.
- **Spin $s > 2$:** No conserved higher-rank tensor currents exist to support non-trivial interactions for massless fields with $s > 2$ in local QFT.